

Min Sung Seo

POSTDOCTORAL SCHOLAR · COGNITIVE NEUROSCIENCE

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Education

Ph.D., Cognitive Psychology, Formal Concentration in Quantitative Psychology

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Aug 2025

- *Advisor:* Kelly S. Giovanello, Ph.D.
- *Thesis:* Prediction Error and Subsequent Memory in Younger and Older Adults

Master of Arts, Cognitive Psychology

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Dec 2023

- *Advisor:* Kelly S. Giovanello, Ph.D.
- *Thesis:* Hippocampal Subfield and Whole-Brain Activity During Mnemonic Discrimination Across the Adult Lifespan

Bachelor of Science, Psychology, Neuroscience Option

THE PENNSYLVANIA STATE UNIVERSITY, UNIVERSITY PARK

May 2020

- *Honors:* Magna Cum Laude

Publications

- **Seo, M.**, Becker, A., Huang, L. H., Overman, A. A & Dennis, N. A. (under review). Spatially unitized associations occupy an intermediate position along the item-to-association continuum in older adults
- West, J. T., Wagner, R. L., Broderick, M., **Seo, M.**, & Dennis, N. A. (under review). Comparing different frontal processes as predictors of older individuals' false memory in a "cage match" between neuropsychological assessments, cortical thickness, neural activation, and representational distinctiveness
- **Seo, M.**, Wagner, R. L., Chamberlain, J. D., Canada, K. L., & Dennis, N. A. (in press). Temporal Interference Modulates Medial Temporal and Frontoparietal Activity During Mnemonic Discrimination: A high-resolution whole-brain fMRI investigation. *Hippocampus*. <https://doi.org/10.1101/2026.03.03.708601>
- **Seo, M.**, Wahlheim, C. N., & Giovanello, K. S. (under review). Stronger mnemonic prediction errors can disproportionately impair memory updating in older adults. *Journal of Experimental Psychology: General*. <https://doi.org/10.31234/osf.io/gcr8a>
- Broderick, M., **Seo, M. S.**, & Dennis, N. A. (in press). The effects of item versus categorical labeling on subsequent memory. *Canadian Journal of Experimental Psychology*
- **Seo, M.**, & Dennis, N. A. (2026). Neural correlates of successful relational memory in younger and older adults: An activation likelihood estimation meta-analysis. *Neuroscience & Biobehavioral Reviews*, 106864. <https://doi.org/10.1016/j.neubiorev.2026.106864>
- Chang, W., Langella, S., **Seo, M.**, Huynh, K., Yap, P., Lin, W., & Giovanello, K. S. (2022). Cross-layer balance of visuo-hippocampal functional connectivity is associated with episodic memory recognition accuracy. *preprint*

Talks

- **Seo, M.** (Jan. 2025). Prediction Error and Subsequent Memory in Younger and Older Adults. *Talk given at the the Cognitive Seminar Series at UNC, Chapel Hill.*
- **Seo, M.** (Oct. 2024). Memory and Aging. *Talk given at the Psi Chi Guest Lecture at UNC, Chapel Hill.*
- **Seo, M.** (Feb. 2024). The Effects of Aging on Prediction Error and Subsequent Memory. *Talk given at the Cognitive Seminar Series at UNC, Chapel Hill.*
- **Seo, M.** (Apr. 2023). Hippocampal subfield and whole-brain activity during mnemonic discrimination across the adult lifespan. *Talk given at the Cognitive Seminar Series at UNC, Chapel Hill.*
- **Seo, M.** (Apr. 2022). Hippocampal subfield activity during mnemonic discrimination across the adult lifespan. *Talk given at the Cognitive Seminar Series at UNC, Chapel Hill.*
- **Seo, M.** (Apr. 2021). Multifeatural encoding in aging. *Talk given at the Cognitive Seminar Series at UNC, Chapel Hill.*

Poster Presentations

- **Seo, M.,** Wahlheim, C. N., Giovanello, K. S. (2025). Mnemonic Prediction Error and Memory Updating in Younger and Older Adults. Poster presented at the Dallas Aging Conference, Dallas, TX.
- **Seo, M.,** Chang, W., Wahlheim, C. N., Giovanello, K. S. (2023). Hippocampal Subfield Activity During Mnemonic Discrimination Across the Adult Lifespan. Poster presented at the North Carolina Cognition Conference, Winston-Salem, NC.
- Chang, W., Langella, S., Huynh, K., **Seo, M.,** Yap, P., Lin. W., Giovanello, K. S. (2022). Investigation of brainwide functional networks associated with hippocampal subfields during memory encoding and retrieval using fMRI with 1-mm isotropic resolution. Poster submitted to the Joint Annual Meeting ISMRM-ESMRMB ISMRT, London, England, UK.
- Gerver, C. R., Overman, A. A., Cowan, J., Jenkins, C., Kautz, B., Long, M., **Seo, M.,** Dennis, N. A. (2020). Manipulating associative encoding strategy impacts neural discriminability at encoding and retrieval. Poster presented virtually at the Cognitive Neuroscience Society Annual Meeting, Boston, MA.

Research Experience

Cognitive Neuroscience of Memory Lab (PI: Kelly S. Giovanello, Ph.D.)

UNC - Chapel Hill

GRADUATE STUDENT

Aug 2020 - Aug 2025

- **Cognitive aging study investigating the effects of aging on prediction error and subsequent memory:** Gathered theoretical background for hypotheses testing, designed a novel experimental paradigm focusing on the role of prediction strength. Currently in prep for manuscript.
- **Lifespan study investigating age related alterations in hippocampal subfield to cortical network functional brain connectivity:** Modified the mnemonic similarity task (MST) for use in whole-brain, high resolution fMRI (3T) to investigate hippocampal subfield function and connectivity with cortical regions during pattern separation processes. Data used for Master's thesis.
- **Hippocampal subfield contributions to associative to associative memory:** Data collection and analyses on an associative memory task for use in whole-brain, high-resolution fMRI (7T) study to investigate hippocampal subfield function and connectivity with cortical regions during (a) discrimination and match detection processes and (b) associative encoding and retrieval.
- **Multifeatural episodic memory encoding in aging:** Data analyses on an existing dataset from a neuroimaging (fMRI) study incorporating a memory encoding task where source features vary independently on two intrinsic dimensions. Age-related differences in whole-brain activation and functional connectivity between ROIs during successful memory encoding were investigated. Manuscript currently under revision.

Cognitive Aging and Neuroimaging Lab (PI: Nancy A. Dennis, Ph.D.)

Penn State

RESEARCH ASSISTANT

Aug 2018 - May 2020

- **Role of unitization in successful encoding of two pieces of information:** Data collection and preprocessing (imaging data) on an associative memory task utilizing a unitizing strategy to test unitization's effect on later memory retrieval in younger and older adults.
- **Influence of shcemas on memory for non-shcematic information in younger and older adults:** Data collection on a novel scene memory task paradigm to investigate age-related differences in the influence of schemas on memory for non-schematic information.
- **Effect of culture in associative memory:** Data collection on an associative memory task to investigate the effect of culture on associative memory.

Center for Language Science (Primary Advisor: Melinda Fricke, Ph.D.)

Penn State

RESEARCH ASSISTANT

Jan 2016 - May 2016

- **Differences in task switching during English-Spanish cognates and non-cognates:** Data collection using EEG and recording of English-Spanish bilinguals to identify differences in neural processes between English-Spanish cognates and non-cognates during L1 - L2 switching.

Honors & Awards

2025	Awardee , The Graduate School Summer Research Fellowship, \$8000	UNC - Chapel Hill
2025	Awardee , Graduate Student Transportation Grant, \$1400	UNC - Chapel Hill
2025	Awardee , Sallie P. Asche Award, Dallas Aging Conference, \$500	UT - Dallas
2025	Departmental Nominee , Dean's Distinguished Dissertation Award	UNC - Chapel Hill
2024	Awardee , Dashiell Dissertation Startup Award, \$1000	UNC - Chapel Hill
2022,2023	Awardee , Dashiell Travel Award, \$1000, \$200	UNC - Chapel Hill
2021	Departmental Nominee , The Graduate Education Advancement Board Impact Award	UNC - Chapel Hill